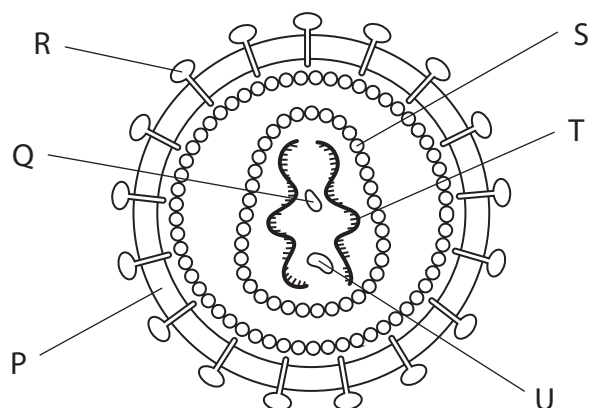


6 The diagram shows a human immunodeficiency virus (HIV).



(a) (i) Which structure is RNA?

(1)

- ☐ **A** R
- ☐ **B** Q
- ☐ **C** S
- ☐ **D** T

(ii) How many of the following structures contain protein?

- structure Q
- structure R
- structure S

(1)

- ☐ **A** 0
- ☐ **B** 1
- ☐ **C** 2
- ☐ **D** 3

(iii) Which virus has structure P and contains RNA?

(1)

- ☐ **A** Ebola virus only
- ☐ **B** Ebola virus and tobacco mosaic virus (TMV)
- ☐ **C** Tobacco mosaic virus (TMV) only
- ☐ **D** Neither Ebola virus nor tobacco mosaic virus (TMV)

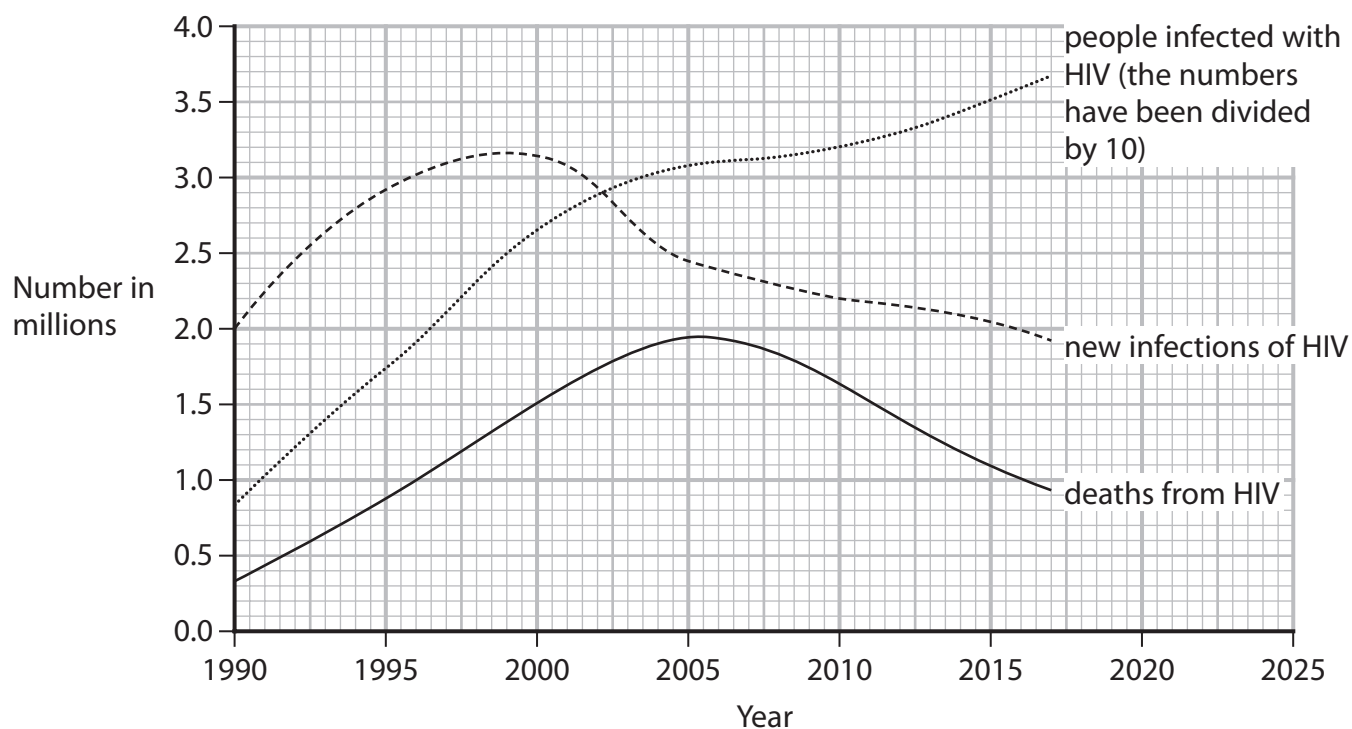
(iv) The diameter of HIV is 120 nm.

Which is the approximate magnification of this drawing?

(1)

- ☐ **A** $\times 500$
- ☐ **B** $\times 5\,000$
- ☐ **C** $\times 50\,000$
- ☐ **D** $\times 500\,000$

- (b) The graph shows the number of HIV infections and the number of deaths from HIV in the world from 1990 to 2017.



- (i) The data plotted for the number of people infected with HIV has been divided by 10.

Explain why these numbers were divided by 10.

(2)

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- (ii) Calculate the mean rate of increase in the number of people infected with HIV from 2005 to 2010.

Express your answer in standard form to an appropriate number of significant figures.

(2)

Answer mean number of infections year⁻¹

- (iii) Explain why the number of people infected with HIV is going up although the number of new infections and deaths is going down, after 2005.

(2)

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- (iv) State how the graph can be used to estimate the number of deaths in 2025.

(1)

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(v) Explain why this estimate would be lower if a vaccine for HIV became available in 2023.

(3)

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(Total for Question 6 = 14 marks)
